

Soil Health and Water Management

Science

May, 2026

Soil Health and Water Management (A32361)

Short Title: Soil and Water Management
Faculty Science and Computing
Department Science
Cluster Agriculture
Author JGERAGHTY
Credits 10

Level: Postgraduate

Description of Module / Aims

The module will review soil biological, chemical, and physical processes and provide demonstration and explanation in field assessment techniques and current laboratory analysis methods. Soil mineral balancing will be investigated and its central role in achieving optimum animal and plant production. Feeding soil to enhance and improve soil biological activity and regenerative agriculture practices for soil nutrient cycling and recycling will also be studied. Regulations pertaining to conversion to organic production and the amendment of organic products to land will be reviewed. The role of soil texture and structure on drainage, soil trafficability, and water availability will be appraised along with the management of soil physical quality. The primary objective of this course is to enable, and empower, participants to assess soil profiles, associated properties and understand how to prevent and mitigate soil degradation and improve overall soil health, function, and water relations for climate resilient food production.

Programmes

SCIE-0080	MSc in Organic and Biological Agriculture (WD_SOBIA_R)	1 / 1 / M
SCIE-0080	Postgraduate Programmes in Science (WD_SGRAD_XX)	5 / 0 / E

Indicative Content

- Soil biological, chemical, and physical field assessment and laboratory analysis methods
- Soil nutrient and mineral balancing to optimise animal and plant production in regenerative agriculture
- Feeding soil for enhanced soil biological activity for improved nutrient cycling and recycling in regenerative agriculture
- Soil fertility maintenance necessary for plant, animal and human health following transition to organic low input systems
- Assessment of soil structure and prevention and mitigation of degradation, compaction, and erosion
- Optimising the water cycle - effective versus non-effective rainfall and water availability and storage
- Harnessing water resources for livestock hydration and maintenance and crop irrigation systems
- Developing policy and regulation for soil and water management in the EU Green Deal, F2F Strategy, and the Programme for Climate Action

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and explain priority field and laboratory practices for soil assessment, analysis methods, and techniques.
2. Interpret soil analyses to estimate optimum inorganic and organic fertiliser levels required to balance the soil nutrient profile for improved output and production.
3. Show how to manipulate soil micro-organism activity to enhance soil biological processes for nutrient cycling and recycling.
4. Categorize and discriminate between different land management practices to reverse soil degradation and improve physical function.
5. Arrange and combine evidence for natural water management practices to develop plans for climate resilient production.
6. Estimate and evaluate the cost-effectiveness and profitability of water irrigation systems and storage practices for crop and livestock production.
7. Critique, interpret and summarize appropriate and relevant scientific research information and associated best practice for climate resilient soil and water management.

Learning and Teaching Methods

- Online lectures, guest lectures, laboratory practical, podcasts
- Field visit and assessment for class discussion
- Self-directed case-study for a farm natural resource audit

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6,7
Continuous Assessment	50%	1,2,3,4,5,6,7
<i>Case Studies</i>	30%	1,2,3,4,5,6,7
<i>Assignment</i>	20%	1,2,3,4,5,7

Assessment Criteria

- <40%:** Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.
- 40%--59%:** The learner will be able to demonstrate a good understanding of the different aspects to soil health and how these can be assessed and measured. The student will be able to conduct and write a case study/resource audit.
- 60%--69%:** In addition to the above, the student will demonstrate more detailed knowledge of soil mineral balancing, and module topics including effective and efficient water storage and utilisation. The student will also be able to write and present a good quality case study on soil and water resources.
- 70%--100%:** The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including an innate understanding that all animal and crop production is founded on the sustainable management of these primary resources and produce a comprehensive audit case study on soil and water resources.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		12
Practical		12
Online Delivery		24
Independent Learning		222

Essential Material

- Albrecht, W.A. and C. Walters . _Albrecht on Soil Balancing_. 2nd ed.. Greeley, Colorado: Acres USA, 2011.
- Albrecht, W.A. and C. Walters. _Soil Management._. Greeley, Colorado: Acres USA, 2011.

Supplementary Material

- Emmett-Booth, J.P. et al. _The Soil Structure ABC_. 1st ed. Dublin: DAFM/Teagasc/UCD, 2018.

Requested Resources

- Room Type: Biology Lab